॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

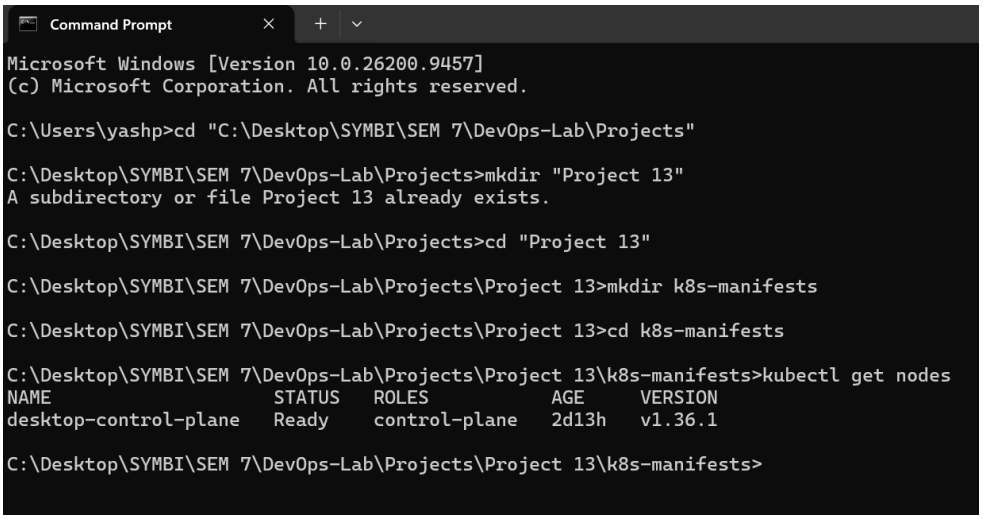
Re-accredited by NAAC with 'A' grade (3.58/4) | Awarded Category – I by UGC

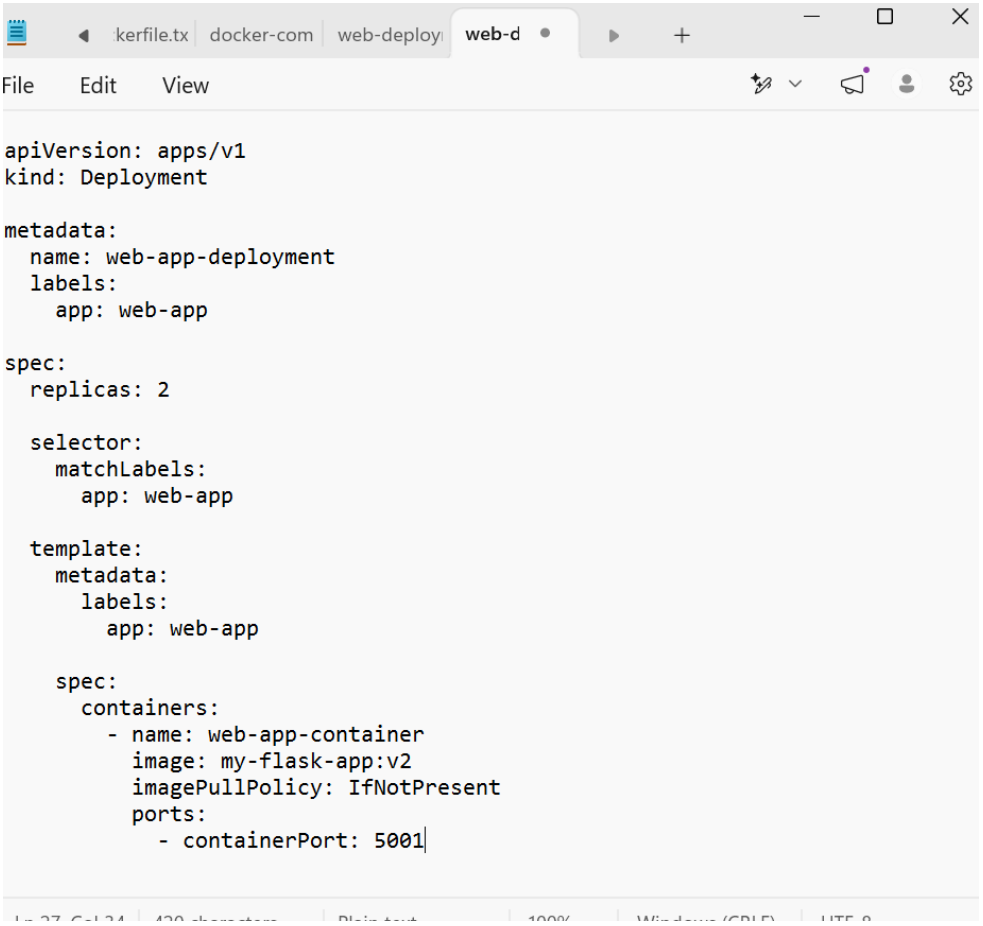
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

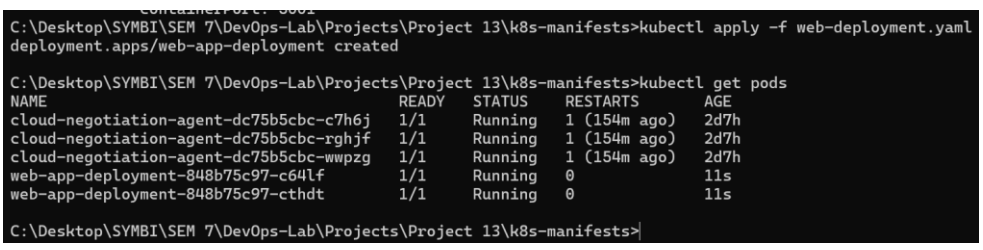
Assignment No. 13

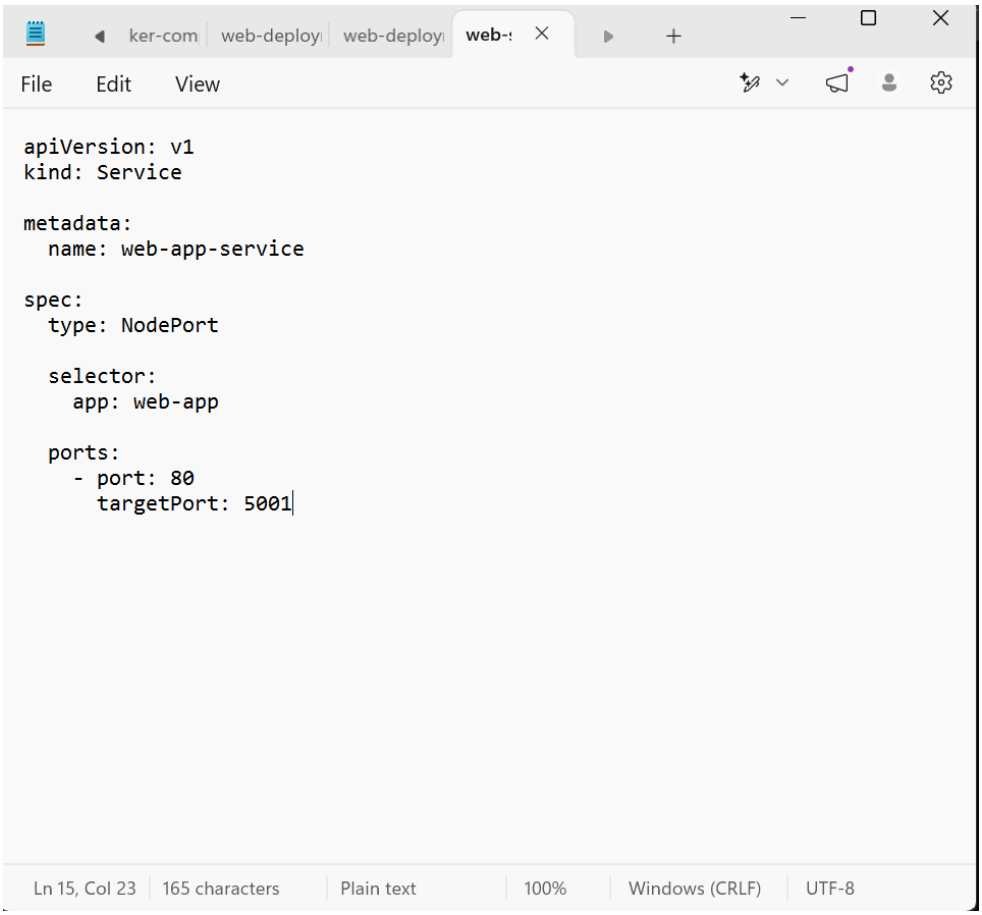
Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Yash Patel**
PRN No.	**23070122235**
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Kubernetes Service & Networking
Theory:	A Kubernetes Service is an abstraction that provides a stable network endpoint for accessing one or more Pods in a Kubernetes cluster. Since Pods are temporary and their IP addresses may change, a Service uses labels and selectors to route incoming traffic to the appropriate Pods. A NodePort Service exposes the application outside the cluster by allocating a port on each cluster node, allowing external access to the application. During local development, kubectl port-forward can also be used to securely forward traffic from a local machine to a Kubernetes Service for testing and verification. This approach simplifies communication between applications and ensures reliable access to containerized workloads. **Commands Used** • Created the Kubernetes Deployment: kubectl apply -f web-deployment.yaml • Verified that the Pods were running: kubectl get pods • Created the Kubernetes Service: kubectl apply -f web-service.yaml • Displayed the available Kubernetes Services: kubectl get service • Displayed detailed information about the Service: kubectl describe service web-app-service • Displayed the Service endpoints: kubectl get endpoints

	• Displayed the running Pods with labels: kubectl get pods --show-labels • Displayed the application logs: kubectl logs web-app-deployment-<pod-name> • Forwarded the Kubernetes Service to the local machine: kubectl port-forward service/web-app-service 8085:80 • Deleted the Kubernetes Service: kubectl delete -f web-service.yaml • Deleted the Kubernetes Deployment: kubectl delete -f web-deployment.yaml
Implementation:	**Step 1: Create the Kubernetes Service YAML** Created Kubernetes Service Manifest — Created the web-service.yaml file defining a Kubernetes NodePort Service to expose the Flask web application running in the Kubernetes Pods. **Step 2: Apply the Servic** Applied Kubernetes Service — Applied the Kubernetes Service manifest using kubectl apply, creating a NodePort Service for the deployed Flask application. **Step 3: Find the Service Information** Verified Kubernetes Service — Verified the Kubernetes NodePort Service and identified the assigned NodePort (30649) used to expose the Flask web application outside the Kubernetes cluster. **Step 4: Access the Application** Accessed Application Through Kubernetes Service — Used kubectl port-forward to forward the Kubernetes Service to the local machine and successfully accessed the Flask web application at http://localhost:8085, verifying that it was running correctly. **Step 5: Clean Up** Deleted Kubernetes Service — Removed the Kubernetes Service using the kubectl delete command, cleaning up the networking resources created for the application.


Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\yashp>cd "C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects"
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>mkdir "Project 13"
A subdirectory or file Project 13 already exists.
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>cd "Project 13"
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13>mkdir k8s-manifests
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13>cd k8s-manifests
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl get nodes
NAME STATUS ROLES AGE VERSION
desktop-control-plane Ready control-plane 2d13h v1.36.1
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>


kerfile.tx
docker-com
web-deploy
web-d
File
Edit
View
apiVersion: apps/v1
kind: Deployment
metadata:
name: web-app-deployment
labels:
app: web-app
spec:
replicas: 2
selector:
matchLabels:
app: web-app
template:
metadata:
labels:
app: web-app
spec:
containers:
- name: web-app-container
image: my-flask-app:v2
imagePullPolicy: IfNotPresent
ports:
- containerPort: 5001


C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl apply -f web-deployment.yaml
deployment.apps/web-app-deployment created
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl get pods
NAME READY STATUS RESTARTS AGE
cloud-negotiation-agent-dc75b5cbc-c7h6j 1/1 Running 1 (154m ago) 2d7h
cloud-negotiation-agent-dc75b5cbc-rghjf 1/1 Running 1 (154m ago) 2d7h
cloud-negotiation-agent-dc75b5cbc-wwpzg 1/1 Running 1 (154m ago) 2d7h
web-app-deployment-848b75c97-c64lf 1/1 Running 0 11s
web-app-deployment-848b75c97-cthdt 1/1 Running 0 11s
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>




```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl apply -f web-service.yaml
service/web-app-service created

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl get service
NAME                        TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
cloud-negotiation-service   NodePort    10.96.136.229   <none>        5000:30500/TCP   2d8h
kubernetes                  ClusterIP   10.96.0.1       <none>        443/TCP          2d14h
web-app-service             NodePort    10.96.221.219   <none>        80:30949/TCP     8s

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl describe service web-app-service
Name:                     web-app-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=web-app
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.96.221.219
IPs:                      10.96.221.219
Port:                     <unset>  80/TCP
TargetPort:               5001/TCP
NodePort:                 <unset>  30949/TCP
Endpoints:                10.244.0.11:5001,10.244.0.10:5001
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl get pods --show-labels
NAME                                    READY   STATUS    RESTARTS      AGE    LABELS
cloud-negotiation-agent-dc75b5cbc-c7h6j   1/1     Running   1 (157m ago)   2d7h   app=cloud-negotiation-agent,pod-templa
te-hash=dc75b5cbc,tier=api,version=v1
cloud-negotiation-agent-dc75b5cbc-rghjf   1/1     Running   1 (157m ago)   2d7h   app=cloud-negotiation-agent,pod-templa
te-hash=dc75b5cbc,tier=api,version=v1
cloud-negotiation-agent-dc75b5cbc-wwpzg   1/1     Running   1 (157m ago)   2d7h   app=cloud-negotiation-agent,pod-templa
te-hash=dc75b5cbc,tier=api,version=v1
web-app-deployment-848b75c97-c64lf        1/1     Running   0              3m2s   app=web-app,pod-template-hash=848b75c9
7
web-app-deployment-848b75c97-cthdt        1/1     Running   0              3m2s   app=web-app,pod-template-hash=848b75c9
7

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl get pods
NAME                                     READY   STATUS    RESTARTS       AGE
cloud-negotiation-agent-dc75b5cbc-c7h6j   1/1     Running   1 (158m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-rghjf   1/1     Running   1 (158m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-wwpzg   1/1     Running   1 (158m ago)   2d7h
web-app-deployment-848b75c97-c64lf        1/1     Running   0              3m50s
web-app-deployment-848b75c97-cthdt        1/1     Running   0              3m50s

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl logs web-app-deployment-7d8f9c6b7f-abc12
error: error from server (NotFound): pods "web-app-deployment-7d8f9c6b7f-abc12" not found in namespace "default"

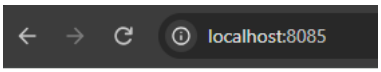
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl logs web-app-deployment-848b75c97-c64lf
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5001
 * Running on http://10.244.0.11:5001
Press CTRL+C to quit

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl port-forward service/web-app-service 8085:80

Forwarding from 127.0.0.1:8085 -> 5001
Forwarding from [::1]:8085 -> 5001
Handling connection for 8085
Handling connection for 8085
```



Hello, Docker! Data has been saved.

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl delete -f web-service.yaml
service "web-app-service" deleted from default namespace

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>kubectl get service
NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
cloud-negotiation-service   NodePort    10.96.136.229   <none>        5000:30500/TCP   2d8h
kubernetes                ClusterIP   10.96.0.1       <none>        443/TCP          2d14h

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 13\k8s-manifests>
```

Subject Faculty Name: Prof. Arijit Dutta